

PRST (11/03/2021)

Markov Inequality

X is a non-negative rv

$$P(X \geq a) \leq \frac{E(X)}{a}$$

where $a > 0$.

proof:

$$I_M(x) = \begin{cases} 1 & \text{if } x \in M \\ 0 & \text{if } x \notin M \end{cases} \quad M \text{ is a set}$$

$$I_M(x) \leq g(x)$$

$$\Rightarrow P_X(M) \leq E[g(x)]$$

$$\left(\begin{array}{l} P_X(M) \\ = P(X \in M) \end{array} \right)$$

$$\left[\begin{array}{l} E(I_M(x)) = 1 \cdot P(X \in M) \\ + 0 \cdot P(X \notin M) \\ = P(X \in M) \end{array} \right]$$

$$M = [a, \infty), \quad g(x) = \frac{x}{a}$$

$$I_M(x) \leq g(x) = \frac{x}{a}$$

$$P(X \geq a) \leq \frac{E(X)}{a}$$

$$\left(\begin{array}{l} \frac{x}{a} > 1 \text{ when } x > a \\ \frac{x}{a} > 0 \text{ when } x < a \end{array} \right) \Rightarrow I_M(x) \leq g(x)$$

Chebyshev Inequality

If X is a rv $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$ ($\sigma^2 > 0$)

$$\text{then } P(|X - \mu| \geq b) \leq \frac{\text{Var}(X)}{b^2}, \quad b > 0.$$

$$\begin{aligned} \text{proof: } P(|X - \mu| \geq b) &= P((X - \mu)^2 \geq b^2) \\ &\leq \frac{E(X - \mu)^2}{b^2} = \frac{\text{Var}(X)}{b^2} \end{aligned}$$

Ex1. Suppose the average grade in a quiz is 70%. Give an upper bound on the proportion of students who scored at least 90% in the quiz.

least 1 goal.

Sol: X - grade in the quiz.

$$E(X) = 0.7$$

$$P(X \geq 0.9) \leq \frac{E(X)}{0.9} = \frac{0.7}{0.9} = \frac{7}{9}$$

Eg. Suppose a coin is flipped 20 times.
Find a bound for the probability that
~~that~~ 'H' will come at least 16 times.

(Using Markov Inequality).

$$P(H) = 0.2$$

Sol:-

$$P(X \geq 16)$$

X - number of heads
in the 20 tosses.

$$n=20 \quad P(X=x) = \binom{20}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \dots, 20$$

$$P(X=5) = \binom{20}{5} p^5 (1-p)^{20-5}$$

$$p = P(H)$$

$$E(X) = \sum_{x=0}^{20} x P(X=x) = 20 \times 0.2$$

$$P(X \geq 16) \leq \frac{E(X)}{16} = \frac{4}{16} = \frac{1}{4}$$

Eg.

Suppose a fair coin is flipped 100 times.
Find a bound on the probability that
number of 'H' will be at least 60
or at most 40.

Sol: X - number of 'H' out of 100 tosses

$$E(X) = 100 \times \frac{1}{2} = 50$$

$$Var(X) = 100 \times \frac{1}{2} \left(1 - \frac{1}{2}\right) = 25$$

$$X \leq 40$$

$$\binom{n}{x}$$

$$P(X > 5)$$

$$\begin{aligned}
 & P(X \geq 60 \cup X \leq 40) \\
 &= P(X - E(X) \geq 60 - E(X) \cup X - E(X) \leq 40 - E(X)) \\
 &= P(X - 50 \geq 10 \cup X - 50 \leq -10) \\
 &= P(|X - 50| \geq 10) \\
 &\leq \frac{\text{Var}(X)}{10^2} = \frac{25}{100} = \frac{1}{4}
 \end{aligned}$$

Ex

X is a rv $E(X) = 5$
 $\text{Var}(X) = 3$

Given, $P(|X - 5| < h) \geq 0.99$
 $h = ?$

sol:—

$$\begin{aligned}
 1 - P(|X - 5| < h) &\geq 1 - 0.99 \\
 \Rightarrow P(|X - 5| \geq h) &\leq 0.01 \\
 P(|X - 5| \geq h) &\leq \frac{\text{Var}(X)}{h^2} = 0.01
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= 3 \\
 \Rightarrow \frac{3}{h^2} &= 0.01
 \end{aligned}$$

transformation of rv

X	$P(X=x)$
-1	0.1
0	0.2
1	0.7

$Y = X^2$
 $g(X) = X^2$

Y	$P(Y=y)$
0	$P(Y=0) = 0.2 = P(X=0)$
1	$P(Y=1) = P(X=1) + P(X=-1)$ $= 0.7 + 0.1 = 0.8$

For cont rv:

~~Q~~ X is rv

$$f_X(x) = f(x)$$

$$Y = g(X)$$

g is one to one function.

$$f_Y(y) = f(g^{-1}(y)) \cdot \left| \frac{d g^{-1}(y)}{dy} \right|$$

Ex $X \sim N(0, 1)$ $e^{-\frac{x^2}{2}}$ $-\infty < x < \infty$

$$f_X(x) = \frac{1}{\sqrt{2\pi}}$$

$$Y = e^X$$

$$g(x) = e^x$$

one to one function.

$$g^{-1}(y) = \ln y$$

$$\left| \frac{d g^{-1}(y)}{dy} \right| = \left| \frac{1}{y} \right| = \frac{1}{y}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(\ln y)^2}{2}} \cdot \frac{1}{y}, \quad y > 0$$

one to

$$y = x^2 \quad g(x) = x^2 \quad \text{not one to one}$$

$$x = \begin{cases} \sqrt{y} & , x \geq 0 \\ -\sqrt{y} & , x < 0 \end{cases}$$

$$\left| \frac{dx}{dy} \right| = \left| \pm \frac{1}{2\sqrt{y}} \right| = \frac{1}{2\sqrt{y}}$$

$$f_y(y) = f(\sqrt{y}) \frac{1}{2\sqrt{y}} + f(-\sqrt{y}) \frac{1}{2\sqrt{y}}, \quad y > 0$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{2\sqrt{y}} e^{-y/2} + \frac{1}{\sqrt{2\pi}} \frac{1}{2\sqrt{y}} e^{-y/2}$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-y/2}, \quad y > 0$$